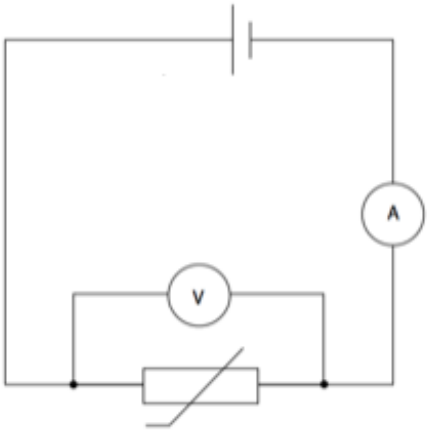


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)			Thermistor symbol (1) Ammeter correct symbol and in series (1) Voltmeter correct symbol and in parallel across the thermistor (1) 		3		3		3
	(b)	(i)		Scale – 1kΩ per cm and 10°C per cm (1) 5 plots correct $\pm$<1 small square (1) Smooth curve between 20 and 100 $^{\circ}$C (1) Ignore below 20 and above 100 when judging line		3		3	3	3
		(ii)		Temperature increases resistance in thermistor decreases (1) at a decreasing rate / less and less (1) Relationship must match the graph		2		2		2

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)	I	8.2 k Ω (+/- 0.2: reading from the candidate's graph) (1) 8.2 = 12 / current (substitution) (1) Current = 12 / 8.2 (Manipulation) (1) = 1.46 / 1.5 mA (1) Manipulation mark can be awarded for current = voltage / resistance $I = V/R$ If 8.4 k Ω is read off graph: Expect answer of 1.43 / 1.4 mA (4) If 8.6 k Ω is read off graph: Expect answer of 1.4 / 1.395 mA (4)		4		4	4	4
			II	0.00146 A (ecf) (1) conversion of candidate answer in part (I) above $\times 12 = 0.0175 / 0.0174W$ (1) Or $(0.00146)^2$ (ecf) $\times 8.2 \times 10^{-3}(1) = 0.0174/0.0175W$ (1) Accept 17.52 for (1)		2		2	2	2
	(c)			Test at lower temperatures (1) obtaining lower temperature using ice (1)			2	2		2
				Question 4 total	0	14	2	16	9	16